

06 — Data Exfiltration

Staging, packing, transfer, rclone, and DNS tunnel evidence

Educational / Defensive DFIR Study Notes

LEARNING OBJECTIVES

- Describe the stage → pack → transfer exfil chain and artifacts at each step
- Use MFT/USN and Hayabusa to time-box staging and tool use
- Prove rclone-based exfil via config, cmdline, and Prefetch
- Recognize DNS tunneling indicators and why adversaries use DNS

SCOPE REMINDER

- This material covers detection, artifacts, and incident response only.
- Attacker actions are phrased as 'adversaries may...' then pivot to evidence.
- Tool/command names appear as IOC strings / exam knowledge — not how-to guides.

Exfil chain

Adversaries may move data in three beats: **stage** (collect into a folder), **pack** (zip/7z/rar, often encrypted/passworded), **transfer** (DNS, HTTP, cloud sync tools like rclone, or other channels). IR should prove each beat when possible.

Phase	Host artifacts	Network artifacts
Stage	New directories; large file creates in MFT; archive tools Prefetch	Internal SMB/copy bursts
Pack	archive.exe / 7z / rar command lines (4688); encrypted containers	Usually still internal
Transfer	rclone/curl/custom beacon cmdline; rclone.conf	Egress to cloud/DNS/HTTP C2; large uploads

MFT & MFTECmd

The Master File Table records file metadata. Using **MFTECmd** (and USN Journal where available) analysts reconstruct create/modify/delete sequences and build timelines of staging folders — even when files were deleted after exfil.

- Look for bursts of file creates in unusual paths (Temp, Public, hidden shares)
- Note delete/rename around the transfer time — cleanup tradecraft
- Combine MFT bodyfile with EVTX for process attribution

Hayabusa triage

Hayabusa turns Windows event logs into a Sigma-oriented timeline. Useful rule categories for exfil-related investigations include:

- Process execution / suspicious command lines
- PowerShell activity
- Service install / scheduled task creation (prep for long-haul exfil)
- Credential dumping detections (often precedes data theft)
- Sigma rule hits stacked by frequency and severity

rclone — detection focus

Adversaries may use **rclone** to push staged data to cloud storage. Defenders prove use via:

Artifact	Location / source	Value
Config file	%APPDATA%\rclone\rclone.conf	Remotes, provider, sometimes token material — isolate carefully
Process create	4688 / Sysmon 1 cmdline	Shows sync/copy and destination remote name
Prefetch	rclone.exe Prefetch on clients	Execution evidence even if binary removed
Network	Proxy/firewall to S3/GDrive/etc. APIs	Volume + timing vs staging

DNS tunneling — detection focus

Adversaries may encode data in DNS queries to bypass HTTP proxies when DNS resolution is allowed outbound. Prove with DNS telemetry + process evidence — not DNS alone.

- Burst of nslookup or unusual resolver clients
- Long, high-entropy, or Base64/Base32-like subdomain labels
- Many unique subdomains under the same parent domain
- Process attribution: which binary issued the queries (Sysmon 22 / EDR / memory netscan)
- Volume far above baseline for that host/user

How do I prove exfil?

- Timeline staging folder via MFTECmd → packing cmdline 4688 → transfer tool cmdline + egress logs
- For rclone: conf + 4688 + Prefetch + cloud provider access logs if available
- For DNS: DNS logs (encoded labels, burst, shared apex) + process that owns queries
- Quantify business impact: which files (names/hashes from MFT) left the environment

Common exam traps

- Calling any ZIP creation 'exfil' without transfer evidence
- Ignoring encrypted archives (still prove stage+pack+egress metadata)
- DNS tunneling conclusions without process attribution
- Forgetting USN/MFT for deleted staging directories
- Missing rclone.conf while hunting only for the EXE name

Lab takeaways

FROM LAB 4-2 — DATA EXFILTRATION

- Separate retained older persistence from new staging by date/time.
- Hayabusa: 4688 row count ≠ Proc Exec event count. Burst rate ≠ DNS packet rate.
- Hash archive members vs staged files; compressed size ≠ source bytes. Reconstruct DNS offline.
- Upload intent: curl + PowerShell history (two sources). Report staged/compressed/received/delivered separately.

QUICK REFERENCE

- Chain: stage → pack → transfer
- MFTECmd for file timeline / deleted staging
- Hayabusa: exec, PS, service, task, cred-dump Sigma
- rclone: %APPDATA%\rclone\rclone.conf + 4688 + Prefetch
- DNS tunnel: bursts, encoded labels, same parent, process proof

Mini Self-Check

Answer these without looking back. Answers are on the next page.

- Q1.** What are the three beats of a typical exfil chain?
- Q2.** Where is the usual user-level rclone configuration path on Windows?
- Q3.** Name two DNS tunneling indicators beyond 'lots of DNS.'
- Q4.** How does MFTECmd help after adversaries delete a staging folder?
- Q5.** Why might adversaries prefer DNS for transfer?

Self-Check Answers

A1. Stage → pack → transfer.

A2. %APPDATA%\rclone\rclone.conf

A3. Long/encoded subdomain labels; many unique labels under one parent; nslookup bursts; mismatch with baseline; tied to a suspicious process.

A4. MFT/USN metadata can still show create/delete/rename history and names/paths for timeline reconstruction.

A5. To bypass HTTP proxies/firewalls when DNS queries are permitted outbound.